



Not all sectors are alike: Differential impacts of shocks in oil prices on the sectors of the Colombian economy

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ABSTRACT

In this paper, an estimation of differential impacts of shocks in oil prices on the economic sectors in the oil-exporting Colombia is made. Using data from the world economy and adopting the model of structural autoregressive vectors (VAR) employed by Kilian et al. (2009), oil prices shocks are estimated according to three types of origin: oil supply, oil demand and aggregate demand. Subsequently, using data from the Colombian economy, the effects that these shocks have on sectoral production in Colombia are estimated through the use of VAR models which allow direct impacts to be separated from those that are transmitted through the interrelations between sectors. The results obtained show that only the specific demand for oil and aggregate demand shocks affect significantly the total production in Colombia, although not in all sectors. Only manufacturing industry, and electricity, gas and water sectors are directly and significantly affected by both types of demand shocks. However, when considering the indirect effects, transmitted through the interrelations between sectors, it was found that the total impacts of these two types of shocks affect the construction, transport and trade, restaurants and hotels sectors. As the effect is positive, the Colombian economy benefits from increases in oil prices due to demand factors, but responds negatively when the opposite situation occurs.

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1. Introduction

The strong increases in real oil prices which were registered at the end of the 1970s and the first half of the 1980s generated an interest in the economic literature to study the effects of unexpected variations in oil prices on the economies of different countries of the world. The seminal works from Darby (1982), Hamilton (1983) and Burbidge and Harrison (1984) coincide in concluding that oil price changes generate a significant impact on the global economic performance.

In recent history, oil price experienced several important changes in its trend. For example, the international oil prices averaged levels close to \$ 20 per barrel during the nineties. However, since 2000, prices faced an upward trend which reached levels above 140 dollars per barrel in the first months of 2008. This was attributed mainly to an increase in world demand and to speculations in the futures market, according to some analysts. In September of the same year, oil prices collapsed by falling below 40 dollars per barrel as a result of the liquidity crisis which was caused in turn by the bankruptcy of Lehman Brothers. As

the world economy recovered from the crisis, oil prices rebounded and remained above 100 dollars per barrel until June 2014. As of that moment, an increase in the oil supply as a consequence of the shale revolution in the United States and a decrease in world aggregate demand led to a price collapse (Prest, 2018), which reached even below 30 dollars per barrel. Since then, prices have recovered slowly but progressively, and they reached around 60 dollars per barrel at the end of 2017.

The behavior of oil prices in the last three decades shows that this variable responds to changes in both supply and demand factors. In line with that, a study from Kilian (2009) evaluated the impact that oil prices shocks had on inflation and economic growth in the US economy between 1973 and 2007. The unexpected changes in oil prices were decomposed in three sources: world oil supply, global aggregate demand and speculative demand for oil. Using a structural autoregressive vector (VAR) model, the author finds that the effects on production and inflation in the United States depend on the source of the changes in oil price. An increase in speculative oil demand decreases real GDP and increases prices in the United States. On the other hand, positive aggregate demand shocks have a slightly positive net effect on production, which implies that the higher world demand direct effect slightly exceeds the negative consequences of a higher oil price. However, both effects translate into higher inflation. In terms of supply, the author found

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that a decline in oil production causes a temporary drop in real GDP and has no major effects on prices (in Kilian's words: "not all shocks are alike").

As in Kilian (2009), most of the previous empirical literature has focused on industrialized and oil-importing countries, such as the United States which is a country in which changes in oil prices have been the main cause of several recession periods and high inflation. However, the effects of these oil prices variations on economic growth vary among countries, especially between an oil-importing and an oil-exporting country. As for the former, an increase in oil prices is generally negative and is considered a negative shock both of supply and demand. In fact, on one hand, it increases transportation costs and product derivatives prices (Kim and Loungani, 1992; Rotemberg and Woodford, 1996; Finn, 2000), and on the other hand affects the terms of trade and disposable income, which in turn affects consumption (Edelstein and Kilian, 2009). For an oil exporting country instead, increases in oil prices generally become a driving factor in their economies, since the demand works inversely because there is an increase in disposable income when oil prices increase. The work of Kilian et al. (2009) and Cashin et al. (2014) show evidence of these differences in the impacts of oil prices changes between oil exporting and importing countries, and it also concludes that results are different depending on whether the variations in oil prices are caused by factors related to world oil production or demand.

Although a greater difference is expected in the response to an oil shock between an oil-exporting country and oil-importing country, the empirical evidence also shows significant differences between two exporting countries (or between two importing countries) in the net impacts of oil price shocks on economic growth. In fact, an outstanding study from Jiménez-Rodríguez and Sanchez (2005) compares oil shock responses in importing and exporting countries. VAR models are used to compare the effects of oil prices shocks on real economic activity in 8 industrialized countries (6 oil-importers and 2 oil-exporter). The authors found that a price increase has a negative impact on 5 of the 6 oil-importing countries analyzed: United States, Germany, France, Italy and Canada. Only in Japan those effects are not significant. As for the two oil-exporting countries analyzed, the results are contradictory: While in Norway an oil price increase has a positive effect on production, in the United Kingdom productive activity experiences a reduction.

The economic structure of these countries might explain these different responses to oil prices changes. In different sectors such as agriculture, construction, manufacturing and transport, oil is an important input in the production process. Therefore, fluctuations in oil prices can affect these sectors, and the effects will depend on the relative importance of oil within the production factors used in each sector. Additionally, there are economic sectors whose demand can be very sensitive to the country's fiscal situation, which in an oil-exporting country can be very dependent on the international prices of this product.

Studies about the disaggregated effects of oil prices shocks on economic sectors exist, but are mainly focused on industrialized countries. Some of the most outstanding works are those of Petersen et al. (1994), Lee and Ni (2002), Jiménez-Rodríguez (2008), Torul and Alper (2010), Arouri (2011), Scholtens and Yurtsever (2012), Shaari et al. (2013) and Jo et al. (2019), in which they all agree in concluding that there are differences in the impacts between different economic sectors within countries.

In the Colombian economy, oil plays a fundamental role in production, exports and fiscal revenues, and thus several studies on the effects of oil prices economy on macroeconomic variables have been carried out. For example, Toro et al. (2015) studied the implications of a negative oil price shock for the Colombian economy. The authors found that the collapse of oil prices affects the terms of trade of the country, impacting also the national income, the current account, the exchange rate, public finances, market confidence and country risk, and generating a significant slowdown in economic activity.

However, only two previous papers investigate the effects of shocks in oil prices on sectoral production in Colombia. Perilla (2010) estimates a structural VAR model which includes oil price, the real exchange rate index and other macroeconomic sectoral variables (production, exports, imports and employment) during the period 1993–2009. This study uses a methodology that is in line with the one used by the majority of international literature. This method estimates independent VAR models for each sector, by replacing the series of aggregate production with the corresponding of the sector of interest. The results reveal that an increase in oil price does not have specific effects on sectoral aggregated value, labor hiring and nor on export and import decisions. On the other hand, Francis and Restrepo (2018) use a different methodology. These authors first estimate a Structural VAR using aggregate series for the economy to identify the oil price shocks that affected Colombia during the period analyzed (2000–2017). Then, the authors use the local projection method to estimate impulse responses of only three sectors (mining, industry and agriculture) to this shock, obtaining mixed results at the sectoral level. The production in the agriculture and mining sectors are unaffected by the shock, while Industry's production falls between the second and fifth quarters after the shock.

The methodological approach employed in both studies does not consider the interrelations between the different economic sectors. Therefore, the estimated effects of oil price shocks on a sector do not take into account the fact that part of the effects can be indirectly transmitted via the impact that other economic sectors have on the sector of interest.¹

Thus, this research seeks to estimate the direct and indirect effects of unexpected oil prices changes over the nine large sectors in which the Gross Domestic Product (GDP) of Colombia can be divided. Importantly, Colombia has an economy in which oil and its derivatives have represented at some point more than 50% of the country's total exports and more than 20% of the current revenues of the national budget.

It is also important to mention that a change in oil prices caused by a global aggregate demand shock may have a different effect than a shock produced by factors affecting oil production worldwide. Therefore, in this paper, the variations in oil prices are differentiated depending on their source, and responses of the economic sectors to the different types of shocks are estimated.

As in Perilla (2010) and most of the previous international studies, structural VAR models are used to determine the sectoral effects of oil price shocks. However, in this study, the sector of interest's production and the other economic sectors's production are included simultaneously in the base model. In addition, restricted versions of this model are estimated in order to "disconnect" the indirect effects, and thus to identify how the transmission of the effects to each sector are carried out. The oil price shocks are first obtained via the VAR model proposed by Kilian (2009). This allows decomposing oil price shocks into shocks of aggregate demand, oil supply and specific demand for oil for speculative reasons.

The rest of the paper is organized as follows: Section 2 describes the methodology used for both obtaining results and estimating the direct and indirect effect of each type of shock on sectoral production in Colombia. In Sections 3 and 4 results are described and discussed, respectively, while in Section 5 the main conclusions are explained.

2. Methods

The methodology used in this work consists of the estimation of two types of Autoregressive Vectors (VAR) models. A first model uses aggregate world demand data and oil production and prices, in order to estimate the oil supply, aggregate demand, and speculative demand for oil

¹ The mentioned work of Jo et al. (2019) is the only known study on the sectoral effects of oil price shocks that takes into account the interrelations that may exist between the different economic sectors. However, its analysis is limited to industrial sectors and fails to make distinctions between direct and indirect effects.

shocks. The second type of model simultaneously incorporates sectors' production information of the Colombian economy and the oil prices shocks which were obtained in the first model to estimate their effects on the sectors' production in Colombia. The estimation of the second type of model is done in two ways: from one side the total effects which oil prices' changes have on the production of the sectors of interest are obtained, from the other side a variant of this model allows to tell apart the effects which are directly or indirectly transmitted to the sector of interest. The latter is done considering the inter-relationships between different economic sectors. Thus, this permits us to overcome the limitations of previous studies for Colombia and in the international literature.

2.1. Estimation of oil price shocks

To obtain the supply and demand shocks which affect oil prices, the methodology used by Kilian (2009) was followed and it consists in estimating a structural VAR model which includes monthly data and in which the vector of endogenous variables is given by:

$$Z_t = [\Delta prod_t \text{ } rea_t \text{ } rpo_t] \quad (1)$$

where $\Delta prod_t$ is the percentage of change in world oil production, rea_t is an index of real economic activity, and rpo_t is the cyclical component of the real oil price.

The dynamics of the autoregressive vector Z_t can be represented in its structural form by the following equation:

$$AZ_t = B(L)Z_{t-1} + e_t \quad (2)$$

where A is a $n \times n$ matrix which describes the contemporaneous relations between variables; $B(L)$ is a $n \times n$ polynomial matrix in the lag operator L , and e_t is a $n \times 1$ vector of structural residuals.

The reduced form that corresponds to this structural model is:

$$Z_t = C(L)Z_{t-1} + u_t \quad (3)$$

where $C(L) = A^{-1}B(L)$ is a polynomial matrix; $u_t = A^{-1}e_t$ describes the relationship between the model residuals in its reduced form and the structural model residuals. Assuming that A^{-1} has a recursive structure, u_t can be decomposed in the following way:

$$u_t \equiv \begin{bmatrix} u_t^{\Delta prod} \\ u_t^{rea} \\ u_t^{rpo} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} e_t^{oil \text{ supply}} \\ e_t^{aggregate \text{ demand}} \\ e_t^{oil \text{ demand}} \end{bmatrix} \quad (4)$$

According to Kilian (2009), oil supply shocks ($e_t^{oil \text{ supply}}$) are defined as unexpected innovations in world oil production, whereas aggregate demand shocks ($e_t^{aggregate \text{ demand}}$) refer to innovations in real world economic activity which do not correlate with oil activity behavior. Finally, innovations in the real price of oil that are not in line with the supply of oil or global aggregate demand are linked to the specific demand for oil. These shocks ($e_t^{oil \text{ demand}}$) are defined as oil prices shocks due to speculative demand (Kilian, 2009) because they do not follow movements in world oil supply or global aggregate demand and they must then reflect changes in demand for precaution or speculation to future behavior uncertainty in the oil market.

The recursive structure of A^{-1} assumes that oil supply does not respond to innovations in oil demand within the same month. This is explained by costs of adjusting oil production which make slow the production response to demand changes. Additionally, the model imposes the exclusion restriction that oil price shocks due to speculative demand do not affect global economic activity within the same month, which is consistent with the slow response that global aggregate demand has historically experienced before changes in oil prices.

For the estimation of the structural VAR model, monthly data were used from January 1975 to December 2017. In line with Kilian (2009),

the estimate was made by ordinary least squares, although 3 lags of the endogenous variables were used. This is the optimal number suggested by the Akaike and Hannan-Quinn information criteria.² Additionally, the Portmanteau residual serial autocorrelation LM test concludes that the hypothesis of no autocorrelation in the residuals cannot be rejected for the first 8 lags with a confidence level of 95%.

The information sources of the different variables are the same used in Kilian (2009). The first difference of the logarithm of the monthly average of crude oil world production in millions of barrels pumped per day was used as an indicator of the growth of world oil production. This information was obtained from the Department of Energy of the United States and was seasonally adjusted using the X12 ARIMA methodology of the Census Bureau of the United States. An index constructed by Kilian (2009) based on dry freight rates was used as a measure of global aggregate demand. This index has been used in other works³ as a proxy for global economic activity as it captures changes in the demand for industrial raw materials in international business markets which are supposed to be highly related to global aggregate demand. The author of this work kept updating the series of this index which is freely accessible for consultation on his personal website, where this information was taken from. The details of the methodology for the construction of the index can be found in Kilian (2009). Finally, the cost of acquisition for refineries of imported crude oil was used as an indicator of the real price of oil. This source comes from the Department of Energy of the United States. This nominal price was deflated using the Consumer Price Index for the United States, available in the FRED database of the Federal Reserve Bank of St. Louis. Subsequently, this series was seasonally adjusted using the X12 ARIMA methodology and its linear trend was estimated to be then subtracted from the seasonally adjusted data and obtain the cyclical component of the real oil price, which was finally the indicator included in the structural VAR.

2.2. Estimation of direct and indirect sectoral effects of oil price shocks.

To estimate the effects of oil price shocks on economic growth in each of the 9 large sectors of the national production, several estimates of the second type of VAR model are built. In this case, the model is built considering the production of the sector of interest and the aggregate of the production of the rest of sectors of the Colombian economy as endogenous variables, whereas lagged oil prices shocks estimated in the previous model are added as exogenous variables. As the shocks in oil prices previously estimated are used as an explanatory variable, and this model is not intended to estimate the structural shocks of the economy, a strategy for the identification of these shocks is not needed. Thus, the VAR is estimated in its reduced form and the ordering of the variables is irrelevant and exclusion restrictions are not established.

The models which are estimated are then represented by the following equation:

$$W_t = F(L)W_{t-1} + G(L)Z_{t-1} + \rho_t \quad (5)$$

where $F(L)$ is a $n \times n$ polynomial matrix in the lag operator L ; $G(L)$ is a $n \times 1$ polynomial matrix in the lag operator L , and ρ_t is a $n \times 1$ vector of model residuals.

The vector of endogenous variables (W_t) is given by:

$$W_t = [prod_{i,t} \text{ } prod_{-i,t}] \quad (6)$$

² The SVAR model estimated by Kilian includes 24 lags of endogenous variables. However, Kilian does not justify the choice of that number of lags. This paper includes 3 lags of the endogenous variables because is the optimal number suggested by the Akaike and Hannan-Quinn information criteria. However, oil price shocks estimated with 3 and 24 lags do not differ much from each other.

³ Apergis and Miller (2009), Alquist and Kilian (2010), Basher et al. (2012), Ratti and Vespignani (2013), and Baumeister and Kilian (2015) are some examples of previous studies which use this measure proposed by Kilian (2009) as an indicator of global economic activity to generally make projections of oil prices.

where $prod_{i,t}$ represents the production of the sector of interest of the national economy, and $prod_{-i,t}$ is the total production of the country minus the production of the sector of interest.

The vector of exogenous variables (Z_{it}) is given by:

$$Z_{it} = [e_{j,t}] \quad (7)$$

where $e_{j,t}$ are the oil prices shocks of interest (either supply, aggregate demand or speculative demand for oil). Eighteen lags are included in these shocks. This is due to the fact that robustness tests show that only from this value an increase in lags number begins to significantly affect the statistical significance of the results. This is because the study period has a limited number of quarters (96, equivalent to 24 years). For the estimation of this model, in fact, quarterly data of the logarithm of seasonally adjusted sectoral production for the period 1994–2017 are used. The latter information was obtained from the National Administrative Department of Statistics (DANE), although it was necessary to splice the series as during the study period three methodologies were implemented in Colombia (one in 1994, one in 2000 and one more in 2005) for the estimation of the country's national accounts.

As shocks in the previous model are monthly obtained, a cumulative quarterly measure of the shocks was constructed which results from the sum of the 3 monthly shocks registered during the quarter. This strategy is basically the same used by Kilian (2009), who made a quarterly average of the monthly shocks in oil prices he had previously estimated. This is to assess the effect of these shocks on economic growth in the United States, for which only quarterly data were available.

The estimation of the base model was made using a lag of the endogenous variables, which is the optimal number suggested by the information criteria Schwarz and Hannan-Quinn. With this specification, the Portmanteau residual serial autocorrelation LM test applied to this base model concludes that the hypothesis of no autocorrelation in the residuals for the first 8 lags with a confidence level of 95% cannot be rejected.

Based on the estimation of this model, the impulse response functions are obtained. They show the total effects generated on the production of each of the sectors in response to the changes in the exogenous variable, that is, in response to the shock in oil prices of interest (supply or aggregate demand or speculative demand).

Note that the estimated responses have the generated regressor issue, as outlined in Pagan (1984) and Murphy and Topel (1985), since oil price shocks are estimated. The OLS estimator for this two-step estimation method yields consistent estimates of the coefficients in the second stage regression, but statistical inferences based on conventional standard errors is biased in favor of rejecting the null hypothesis. To account for that issue, standard errors are computed by bootstrap methods as in Romer and Romer (2004 and 2010) and Ashraf and Galor (2013). Standard errors were calculated by making 10,000 estimates of the vector of coefficients obtained from a multivariate normal distribution. Mean, variance and covariance matrix are equal to the point estimators and the variance and covariance matrix of the coefficients of the estimated regressions.

The results obtained from the estimation of this model for each sector of interest do not consider that, with high probability, a significant part of the impacts of oil price shocks on a particular sector will be received indirectly by the sector through interaction with other sectors of the economy. Therefore, in order to separate the direct effects from the indirect effects, a variant in the estimation of the base model is made and includes the variable $prod_{-i,t}$ as an exogenous variable in the estimation of each sectoral VAR. This procedure is equivalent to zeroing the coefficients of the VAR model equation in which the dependent variable is the variable indicator of the production of the other economic sectors and which “disconnects” the transmission of the oil prices shocks effects on a sector of interest through the impact received by the other sectors of the economy. This strategy has been adapted from the implementation carried out by Ramey (1993), Endut et al. (2013) and

Quintero (2015) to determine the relative importance of certain channels in the transmission of monetary policy to real activity.

The vectors of endogenous variables (W_t) and exogenous variables (Z_t) in the estimation of the direct effects of the shocks in oil prices on the economic sectors are then determined by the following equations:

$$W_t = [prod_{i,t}] \quad (6')$$

$$Z_{it} = [prod_{-i,t} e_{j,t}] \quad (7')$$

Indirect impacts are then obtained as the difference between the impulse response functions of the base model and the restricted model.

Additionally, in order to have a reference of the impact magnitude generated on the total production in Colombia, a VAR model similar to the previous one is also estimated, in which the production of the aggregate is included as the only endogenous variable of the country's economy and the exogenous variables are the shock in oil prices of interest. That is, the vectors of endogenous variables (W_t) and exogenous variables (Z_t) are given by:

$$W_t = [totalprod_t] \quad (6'')$$

$$Z_{it} = [e_{j,t}] \quad (7'')$$

where $totalprod_t$ represents the production of the total Colombian economy.

3. Results

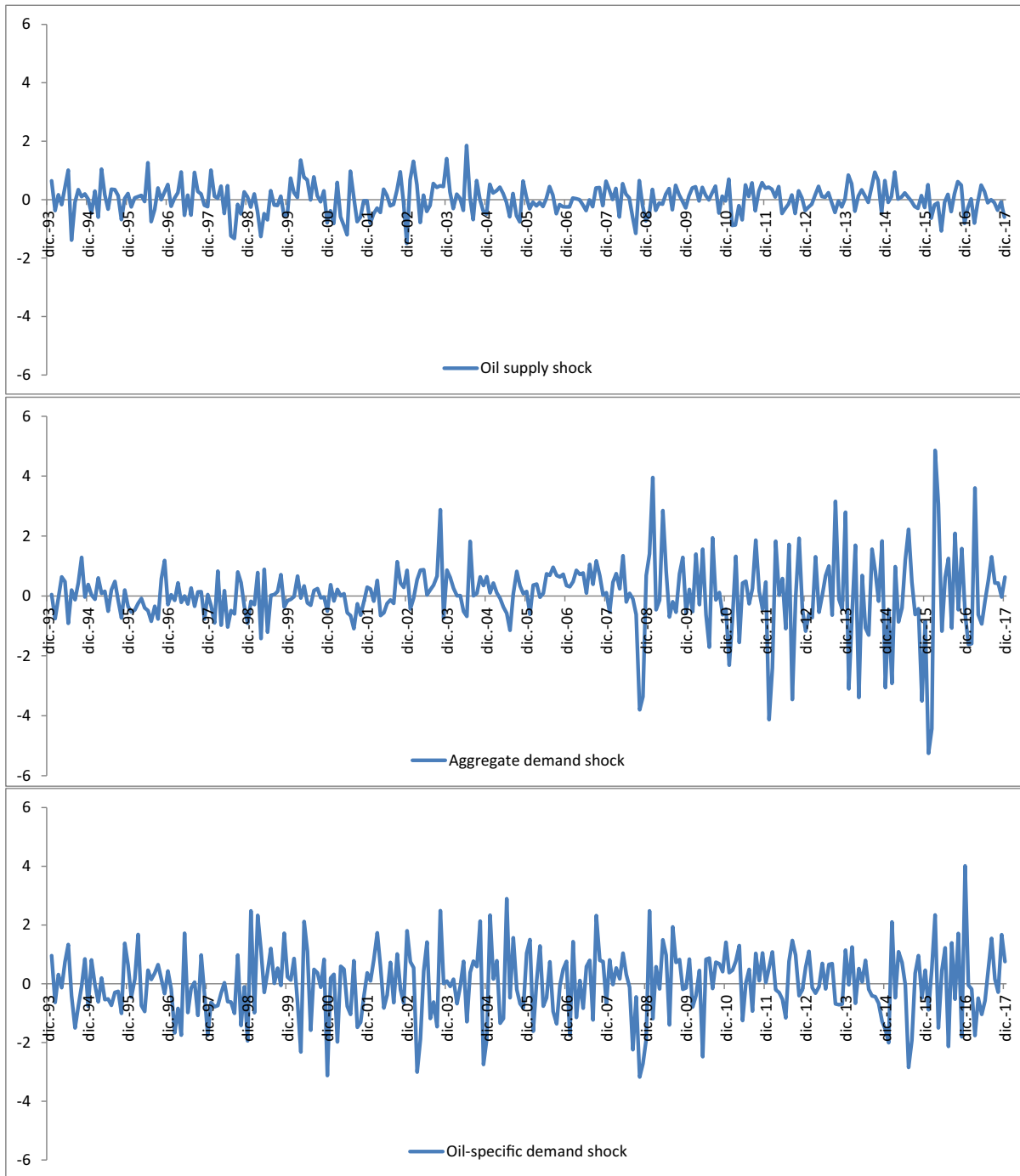
3.1. Oil price shocks

Based on the estimation of the structural VAR model described in Section 2.1, the oil supply, aggregate demand and specific demand for oil shocks were obtained. Although the estimation period of this first model was from January 1975 to December 2017, Fig. 1 shows the resulting shocks for the period from January 1994 to December 2017, in which the impact is subsequently evaluated on sectoral economic growth. Additionally, Fig. 2 shows the contribution of each shock to the real oil price for the same period, based on the historical decomposition of the shocks.

As the methodology, the variables and the source of information are the same as the ones in Kilian (2009), the results in this study of the behavior of the shocks and the contribution of each of them to the real price of oil until 2007 is very similar to what obtained by this author. It is worth noting that in the mid-1990s, negative shocks of aggregate demand and speculative demand for oil were generated, which had a significant impact on oil prices. These negative shocks deepen in 1997 and 1998 as a result of the Asian crisis and also end up having an important impact on oil price, which reaches historic lows, both in nominal and real terms.

As of 1999, oil prices recovered and this is explained by demand factors, especially by specific demand for oil. However, this recovery is interrupted by the 2001 US recession, which generates negative shocks of aggregate demand for a short duration. As a result of China's growth, strong and prolonged positive demand shocks generated between 2002 and 2007, especially of aggregate demand which have an important impact on oil prices (Fig. 2).

In the period 2008–2017, positive supply shocks exerted some downward pressure on oil prices. However, this greater world oil production has not been the main determinant of oil prices shocks, nor have they been since the 70s, as Kilian (2009) showed. After reaching levels above 100 dollars per barrel in 2008, that same year the oil price fell sharply as a result of the financial crisis that began in the United States, becoming a strong shock both in aggregate demand and in specific demand for oil. As the world economy recovers, shocks of aggregate demand and speculative demand for oil generated and ended



Source: Author's estimation

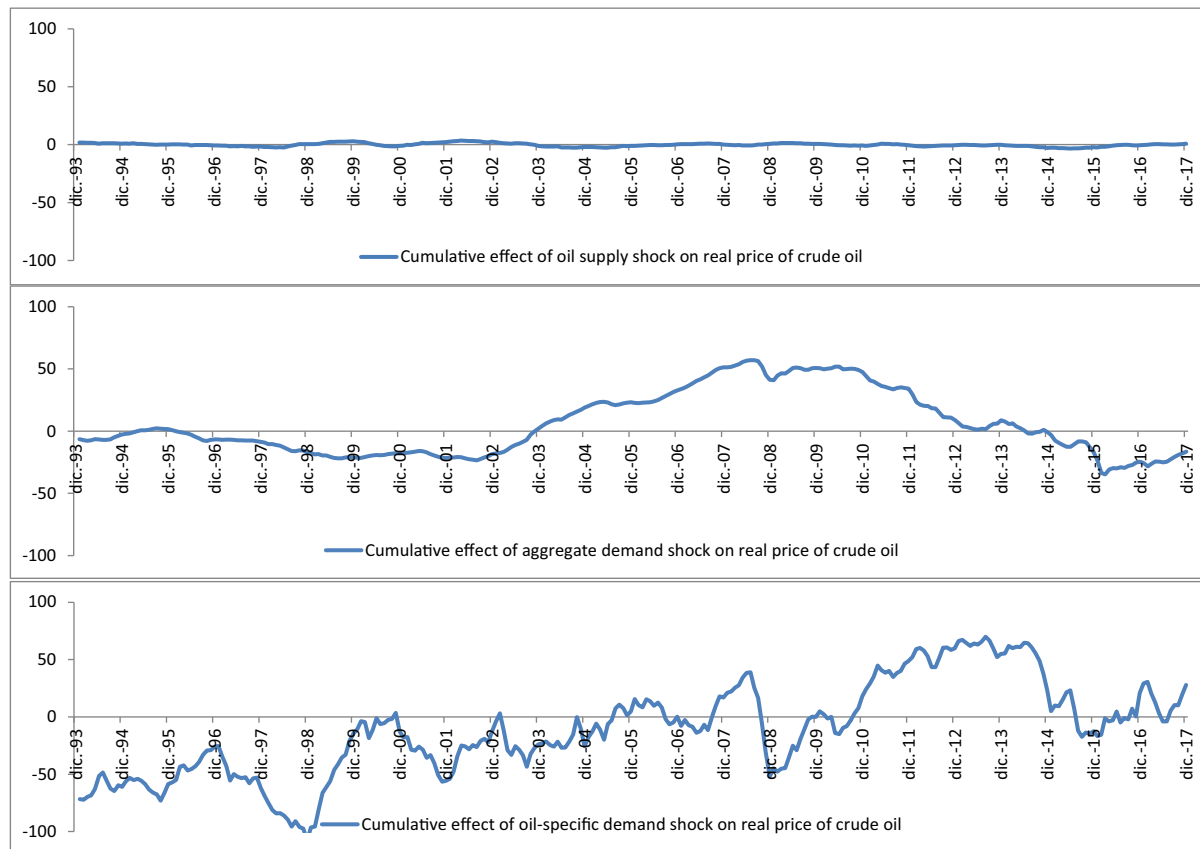
Fig. 1. Structural shocks in oil prices from January 1994 to December 2017.
Source: Author's estimation.

up having a significant impact on the high oil prices until the first half of 2014.

In June 2014, a collapse in oil prices began. This was attributed by some analysts to an excess in the oil supply which in turn weakened the aggregate demand. However, the reality is that the fall in the oil price in the period 2014–2015 is more the result of a market speculative reaction to potential future imbalances than the consequence of a significant and surprising change in market fundamentals. In 2016 and 2017 a slow but progressive recovery in oil prices was reached as a result of production cuts (agreed by the OPEC), difficulties in Venezuela's production and a rebound in global aggregate demand driven mainly by

China and India. Moreover, also speculative factors that had a significant impact on the recovery of oil prices played a role, especially those related to geopolitical tensions in the Middle East.

In terms of the impacts of the different shocks on oil price over a 4-year horizon, positive shocks in speculative oil demand generate a greater and more sustained increase in the price of oil than the aggregate demand shocks (Fig. 3). A shock of a standard deviation in the speculative demand for oil generates an accumulated impact on the price of oil to levels close to 9%. Instead, the maximum impacts of an aggregate demand shock reach levels slightly above 3%, although this result is statistically significant. On the other hand, positive shocks in the supply of



Source: Author's estimation

Fig. 2. Historical decomposition of oil prices shocks from January 1994 to December 2017.
Source: Author's estimation.

oil generate a decrease in oil prices, but the impacts are of low magnitude (-0.9% at the time of greatest cumulative impact) and they are not statistically significant.

3.2. Impacts of shocks in oil prices on aggregate production in Colombia

As a preamble to the presentation of the results obtained on the sectoral effects of oil price shocks, this section aims to present the results of the impulse response function that shows the impact of aggregate demand shocks, specific oil demand and supply of oil over total production in Colombia. The results were obtained from the estimation of the VAR model in which the only endogenous variable is the production of the aggregate of the country's economy and the exogenous variables are the lags of the oil prices of interest described at the end of Section 2.2. These abovementioned impacts along with the 90% confidence intervals are presented in Fig. 4. From these results, it can be observed that the specific demand for oil and aggregate demand shocks generate a statistically significant impact on GDP in Colombia. In fact, the increases in oil prices due to specific demand shocks have an accumulated impact of 1.5% at the end of the first year, 2.7% in the second year, 3.3% in the third year and 4.2% to the fourth year.

On the other hand, a 1% increase in the real price of oil due to an unexpected increase in world economic activity – or aggregate demand shock – generates a positive impact on national production of 0.9% in the first year, and an 2% accumulated impact after 4 years after the shocks were recorded.

Shocks in world oil production (supply shocks) instead, do not have a significant impact on Colombia's economic growth. A shock of positive oil supply of 1% (decrease in the price of oil of 1% due to an increase in world production) generates, as expected, a decrease in national

production from the first year, but these effects are never statistically significant.

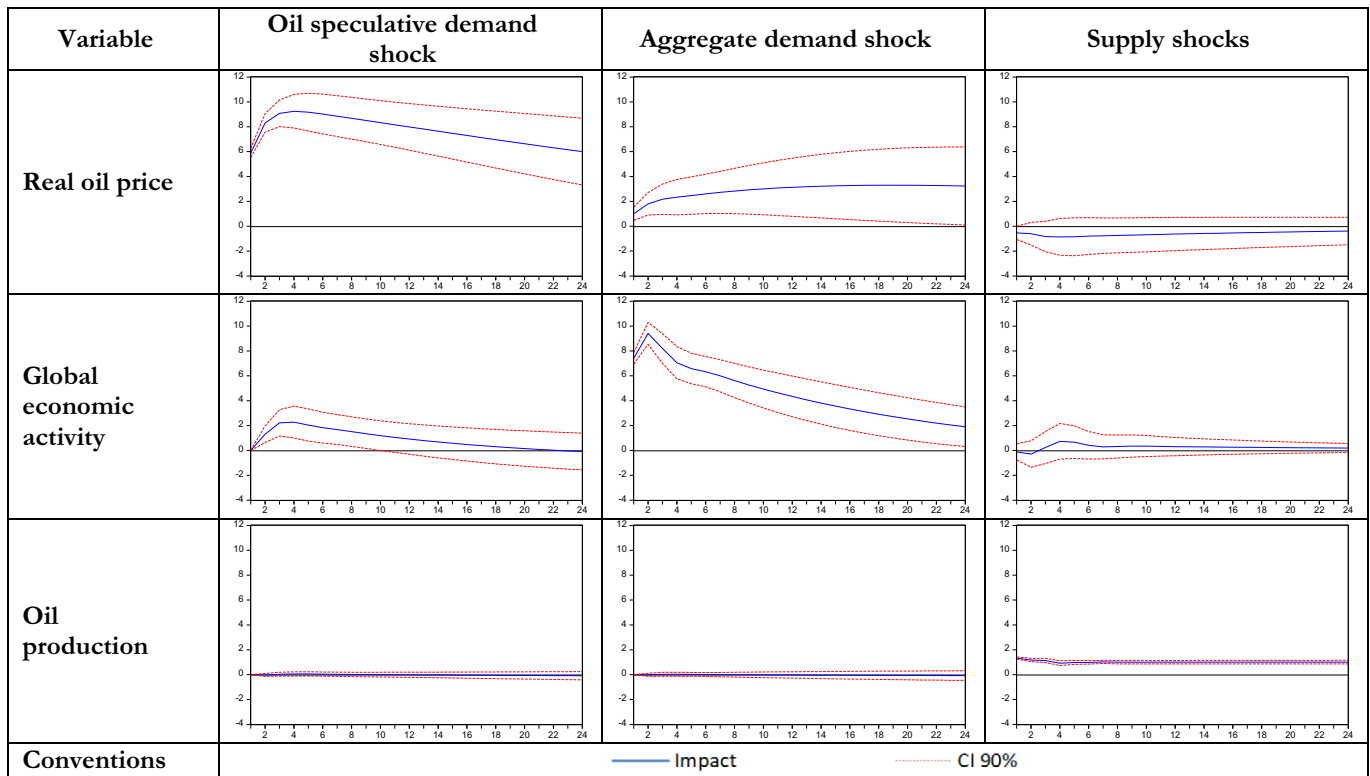
3.3. Impacts of shocks in oil prices on sectoral production in Colombia

Next, the impact of the different types of shocks in oil prices on the production of each of the 9 major sectors of the Colombian economy is presented. The impulse response functions presented in this section are those that result from the estimation of the VAR model for the Colombian economy described in Section 2.2, in which the production of the sector of interest and the consolidated production of the rest of the sector are separately included.

3.3.1. Impact of specific oil demand shocks on sectoral production

According to the results presented in Section 3.2, the unexpected variations in oil prices due to changes in the specific demand for oil are the shocks that generate a more cumulative impact on the total production of the country after 4 years that the shocks occurred. Therefore, this section acquires special relevance when seeking to identify the sectors which are most affected by this type of shocks. The impacts on the nine sectors are presented in Fig. 5. For each type of shocks, the total impact they generate on the production of each sector is presented up to 16 quarters (4 years) after the shocks, along with 90% confidence intervals. Additionally, the figure shows the direct and indirect impacts these shocks have on each sector.

The impulse response functions presented in Fig. 5 show that the speculative oil demand shocks have the greatest effects directly in the manufacturing, electricity, gas and water sectors. After 4 years of a 1% shock in oil price, the manufacturing industry sector reached an accumulated growth of 4.9%, while the electricity, gas and water of 3.8%. In



Source: Author's estimation

Fig. 3. Responses to structural shocks of one standard deviation.
Source: Author's estimation.

both sectors, the total impacts are not very different (in fact, they are slightly lower), which shows that the impacts transmitted indirectly to these two sectors through the other sectors of the economy are not very strong.

However, the greatest total impact as a result of a speculative oil demand shock is observed in the sector of construction, with an accumulated impact at the end of the fourth year of 8%; and in the sector of transportation, storage and communications, trade, restaurants and hotels, and financial intermediation, with impacts of between 4% and 5% four years after the shocks occurred. However, it is striking that in all these sectors the impacts are not received directly but rather indirectly through their relationship with the rest of the sectors of the economy. In all these sectors the effect is positive, which means that the impact of an increase in demand for oil for speculative reasons, which increases international oil prices, generates an expansion in these three sectors.

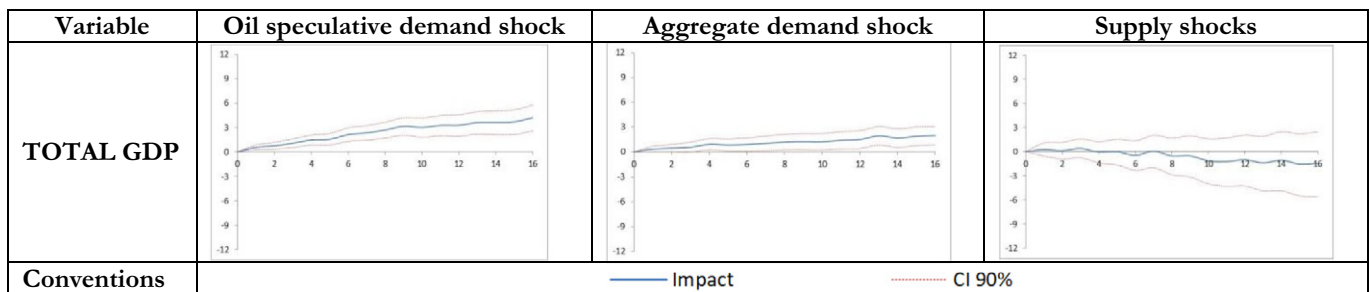
In the other three sectors (agriculture, mining and services) the direct impacts are not statistically significant, as they do not receive a

very strong transmission of the impacts generated in the rest of the sectors of the economy.

3.3.2. Impact of aggregate demand shocks on sectoral production

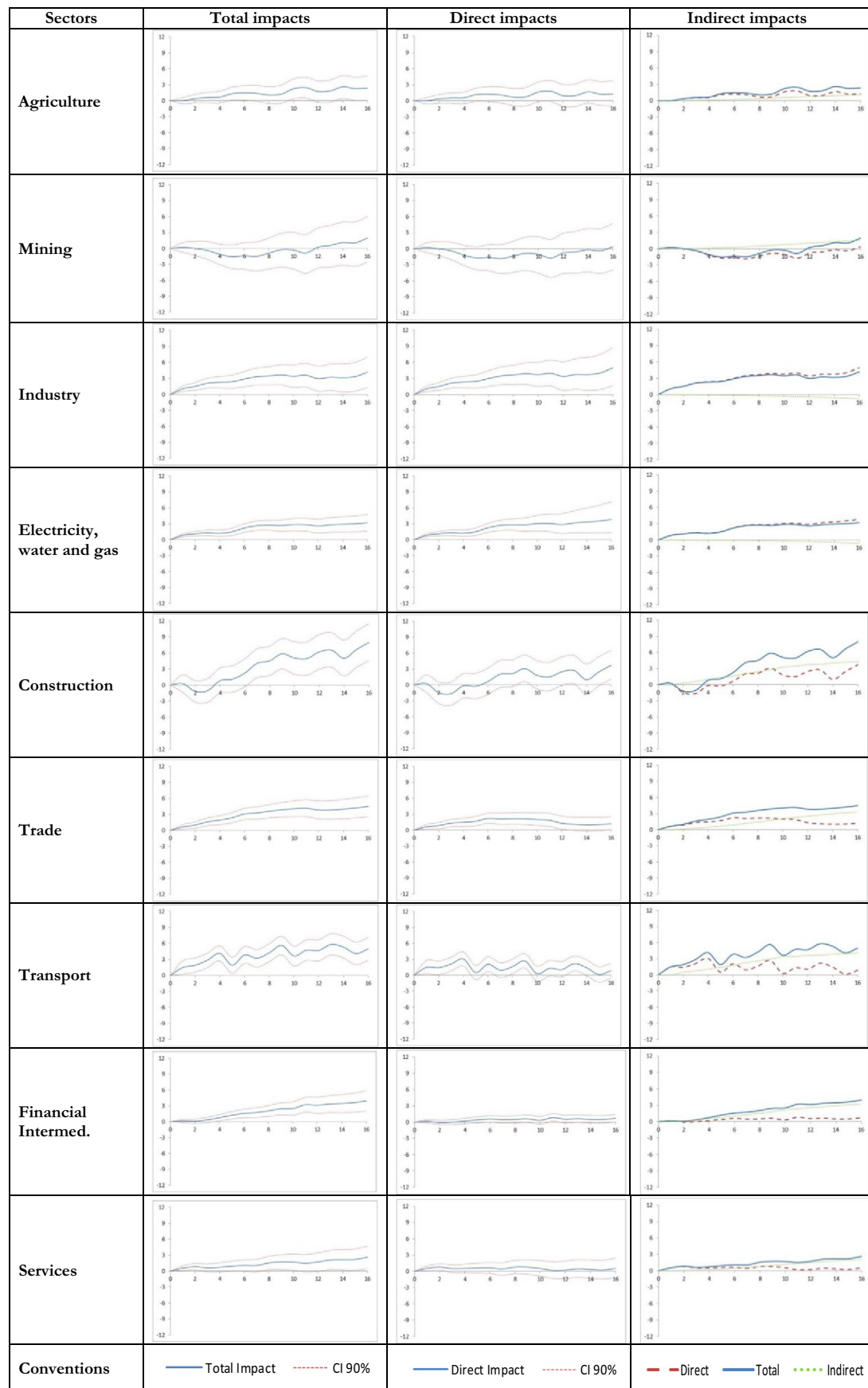
Aggregate demand shocks – similarly to shocks of specific demand for oil – also showed significant impacts on Colombian production aggregate. Moreover, the sectoral analysis revealed that an increase in oil price due to an increase in aggregate demand affected those sectors which respond the most to oil specific demand shocks.

In fact, manufacturing and electricity, gas and water industries are the sectors with the greatest direct effects, as can be seen in Fig. 6. In both sectors, the maximum impact is shown three years after the shocks occurred. In the first sector, the maximum impact is 2%, while in the second sector the impact is 1.8%. These impacts are significantly lower than those generated by specific demand shocks for oil, but both types of shocks generated minimal indirect effects in these two sectors.



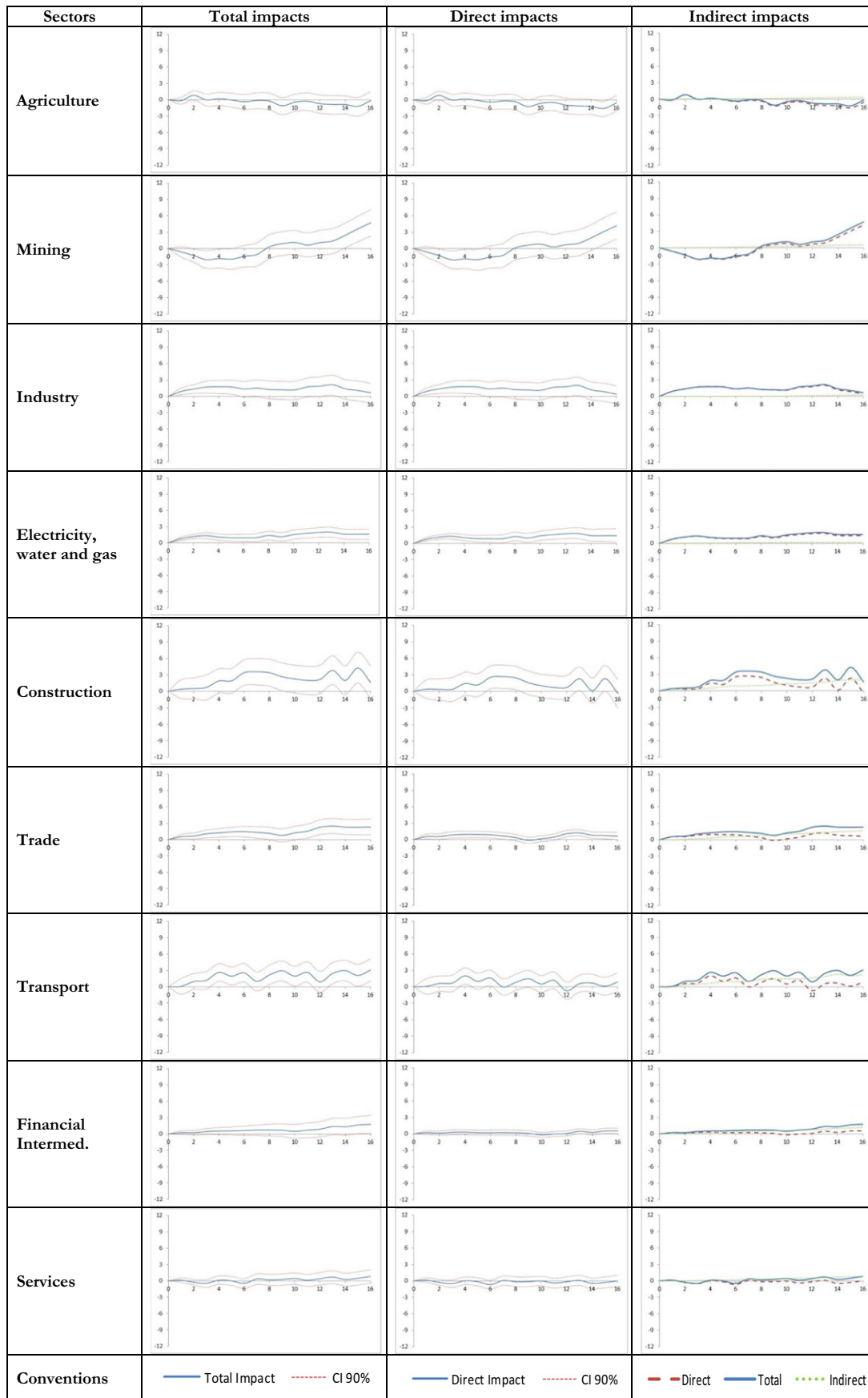
Source: Author's estimation

Fig. 4. Impact of oil price shocks on total Colombian production.
Source: Author's estimation.



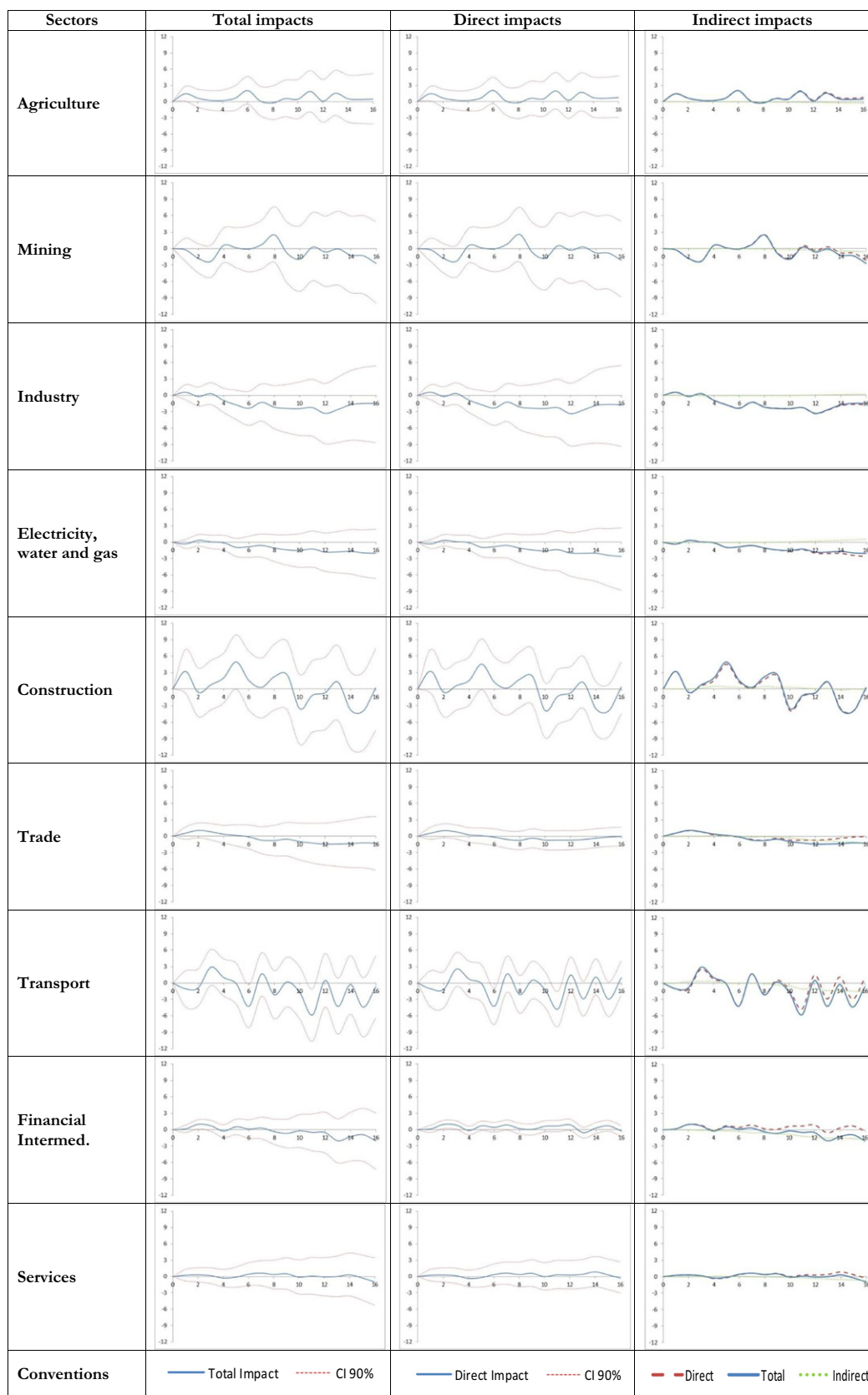
Source: Author's estimation

Fig. 5. Total, direct and indirect impacts of speculative demand shocks on sectoral production in Colombia.
Source: Author's estimation.



Source: Author's estimation

Fig. 6. Total, direct and indirect impacts of aggregate demand shocks on sectorial production in Colombia.
Source: Author's estimation.



Source: Author's estimation

Fig. 7. Total, direct and indirect impacts of supply shocks on sectoral production in Colombia.
Source: Author's estimation.

When the impacts transmitted through interaction with the other sectors of the economy are also considered, it is found that the construction sector is the most positively affected by an aggregate demand shock, although the maximum impact accumulated in the four years following the occurrence of shocks is 4.3%. It is also found again that the sector of transport, storage and communications, and sector of trade, restaurants and hotels have significant total impacts (3.1% and 2.5% respectively in maximum effect period). Unlike what happened with specific demand shocks for oil, the financial intermediation sector does not have a direct or indirect impact on the aggregate demand shocks.

Among agriculture, mining and services sectors only mining shows a significant impact during the fourth year after the occurrence of shocks, which is the result of a directly experienced impact.

3.3.3. Impact of global oil supply shocks on sectoral production

Unlike what happens with demand shocks, both aggregate and specific for speculative reasons, none of the sectors of the Colombian economy responds significantly, either directly or fully, to changes in the world oil supply (Fig. 7). These results are consistent with the impact which is generated on the total production of the country which was not statistically significant (Fig. 4).

Although the effects of these shocks are not statistically significant, the largest direct impacts are on manufacturing industry and electricity, gas and water sectors. Here the impacts are negative because a positive shock of supply (increase in world production) reduces the real price of oil, as shown in Fig. 3.

4. Discussion

This paper found that the shocks of specific demand for oil and aggregate demand are those that significantly impact GDP in Colombia. The cumulative impact after 4 years is higher for specific demand shocks than aggregate demand for oil shocks because oil price responds more strongly to speculative demand for oil shocks.

The impacts of the shocks are positive, which means that the increase in oil prices generates a clearly expansive effect on production which is unquestionably associated with higher revenues and greater purchasing capacity of the economic agents, as a result of the production and export of oil.

As per the sectoral effects of these demand shocks, five (construction; transportation, storage and communications; trade, restaurants and hotels; manufacturing; and electricity, gas and water) out of nine sectors in which DANE divides production in Colombia, receive statistically significant impacts from shocks of speculative demand for oil and aggregate demand. Additionally, the financial intermediation sector is significantly impacted by speculative oil demand shocks, but not from aggregate demand shocks.

These sectoral differences in terms of impacts of production from shocks in oil prices can be explained by the exchange rate channel. According to the theory of the Dutch disease, an increase in oil prices generates an appreciation of the exchange rate, which negatively affects the

performance of the exporting sectors. However, this exchange rate appreciation ends up favoring sectors whose products are not tradable but do require a significant amount of imported inputs. This is the concrete case of the construction sector, which receives the greatest impacts in the face of demand shocks.

The results obtained can be also explained by factors that differently affect the demand for goods or services produced in different sectors. One of these factors is the country's fiscal revenues, which in an oil-exporting economy such as Colombia's are directly affected by changes in oil prices. In turn, these changes in the country's fiscal revenues affect certain sectors more strongly than others. Indeed, in the last 15 years, the construction sector has been one of the main drivers of economic activity in Colombia and has been behaving at the same pace as the growth of the economy and tax revenues. In periods of higher fiscal revenues for the country due to higher oil prices, this sector has been favored by the execution of large infrastructure projects such as the fourth generation highway program (4G) and the granting of subsidies by the National Government to the middle and lower class homes for purchasing new housing.

The negative side of the previously described results is that the Colombian economy is negatively affected in a way that a decrease in aggregate demand or the specific demand for oil is experienced. Moreover, the most affected sectors after an oil price reduction are those which benefit the most when shocks are favorable. In fact, in the periods of deceleration of the aggregate economy and decrease in fiscal revenues, the construction sector has experienced strong contractions in Colombia.

On the other hand, it is important to make a distinction from direct and total effects of the impacts obtained in the face of demand shocks. When only direct effects were considered, the manufacturing industry and electricity, gas and water are the sectors that receive the greatest impacts. However, when estimating the total effects which also consider the impacts transmitted through the interrelationships between the sectors, the construction, transportation, storage and communications and trade, restaurants and hotels sectors receive total impacts stronger than the two sectors with the greatest direct impacts.

This is because the impact of the demand shocks on the sectors of the manufacturing industry, and electricity, gas and water, received through the rest of the sectors of the economy is practically null. In principle, it is striking that a sector such as the manufacturing industry does not receive significant stimuli or effects as a result of the effects that the shocks in oil prices have on other sectors of the economy. However, the explanation for these results is found in the matrix of technical coefficients and the matrix of coefficients of total requirements (direct and indirect) of the nine sectors into which the Colombian economy has been divided in this work.

Tables 1 and 2 show the estimation of these matrices in 2000 and 2015, for which the data from the input-output matrices generated by the DANE National Accounts system in Colombia were used. In this way, we have a measure of the direct and indirect productive linkages in Colombia at two different moments of the study period analyzed in

Table 1
Matrix of technical coefficients of Colombian economy - 2000 and 2015.
Source: Author's estimation - DANE - National accounts.

Sector	2000									2015								
	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
1. Agriculture	0.07	0.00	0.15	0.00	0.01	0.03	0.00	0.00	0.00	0.06	0.00	0.13	0.00	0.01	0.03	0.00	0.00	0.00
2. Mining	0.00	0.09	0.03	0.05	0.03	0.00	0.00	0.00	0.00	0.00	0.07	0.04	0.07	0.03	0.00	0.00	0.00	0.00
3. Industry	0.19	0.03	0.34	0.03	0.42	0.18	0.18	0.03	0.10	0.21	0.03	0.36	0.02	0.34	0.18	0.23	0.02	0.08
4. Elec, gas, water	0.00	0.01	0.02	0.24	0.00	0.01	0.01	0.01	0.01	0.00	0.01	0.02	0.29	0.00	0.02	0.02	0.01	0.02
5. Construction	0.00	0.01	0.00	0.04	0.02	0.00	0.01	0.01	0.03	0.00	0.01	0.00	0.03	0.01	0.00	0.00	0.01	0.02
6. Trade	0.00	0.01	0.01	0.00	0.00	0.02	0.07	0.01	0.02	0.00	0.01	0.01	0.01	0.00	0.02	0.06	0.01	0.02
7. Transport	0.01	0.08	0.03	0.01	0.01	0.05	0.10	0.03	0.02	0.01	0.14	0.03	0.01	0.01	0.05	0.09	0.03	0.03
8. Financial Intermed.	0.02	0.02	0.10	0.11	0.08	0.11	0.12	0.13	0.13	0.03	0.03	0.09	0.10	0.10	0.13	0.13	0.17	0.15
9. Services	0.00	0.00	0.01	0.00	0.00	0.01	0.01	0.00	0.07	0.00	0.00	0.01	0.00	0.00	0.01	0.01	0.00	0.08

Table 2

Matrix of total requirements of Colombian economy - 2000 and 2015.

Source: Author's estimation - DANE - National accounts.

Sector	2000									2015								
	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9
1. Agriculture	1.14	0.02	0.26	0.02	0.13	0.09	0.06	0.01	0.04	1.12	0.02	0.23	0.02	0.09	0.08	0.07	0.01	0.03
2. Mining	0.01	1.10	0.06	0.08	0.06	0.01	0.02	0.00	0.01	0.02	1.08	0.07	0.11	0.06	0.02	0.02	0.00	0.01
3. Industry	0.35	0.10	1.64	0.12	0.72	0.35	0.37	0.08	0.22	0.38	0.14	1.69	0.12	0.60	0.35	0.46	0.07	0.20
4. Elec, gas, water	0.01	0.01	0.04	1.33	0.02	0.03	0.03	0.01	0.03	0.02	0.02	0.06	1.41	0.03	0.04	0.05	0.01	0.04
5. Construction	0.01	0.01	0.01	0.06	1.02	0.01	0.01	0.02	0.03	0.01	0.02	0.01	0.05	1.02	0.00	0.01	0.01	0.02
6. Trade	0.01	0.02	0.03	0.01	0.02	1.04	0.09	0.01	0.03	0.01	0.03	0.03	0.01	0.02	1.04	0.07	0.02	0.04
7. Transport	0.03	0.10	0.07	0.03	0.05	0.08	1.13	0.04	0.05	0.03	0.17	0.08	0.04	0.05	0.08	1.13	0.05	0.05
8. Financial Intermed.	0.08	0.06	0.21	0.19	0.19	0.19	0.22	1.17	0.21	0.10	0.10	0.22	0.21	0.20	0.22	0.25	1.22	0.24
9. Services	0.01	0.00	0.02	0.01	0.01	0.02	0.02	0.01	1.08	0.01	0.00	0.02	0.00	0.01	0.01	0.02	0.01	1.09

this work: one in the first half of the study period (2000), and another in the second half (2015).

It is important to have clarity of the interpretation that should be given to the matrices of technical coefficients and total requirements. The matrix of technical coefficients shows the mix of direct inputs that each sector requires from different sectors of the economy. For its part, the matrix of coefficients of total requirements, also called the inverse of the Leontief matrix, shows for each sector not only the inputs required directly from all sectors for production, but also those from the indirect demands of other sectors that provide inputs to the production of the sector of interest.

The results of the matrix of technical coefficients for the years 2000 and 2005 show that in the case of the manufacturing industry there is a high need for inputs from the same sector as the technical coefficients are of 0.34 and 0.36, respectively. A similar situation occurs in the electricity, gas and water sector whose requirements are 0.24 and 0.29, respectively. In no other sector of the economy, inputs are needed higher than 0.13 in the year 2000 and 0.17 in the year 2015. On the other hand, analyzing the results of the matrix of total requirements, in the manufacturing industry and the electricity, gas and water sector, a high amount of input is required from the same sector to produce \$1 of the final product. In effect, to produce \$1 in the manufacturing industry it is required a total amount of inputs (direct and indirect) from the same sector for \$ 1.64 in the year 2000 and \$ 1.69 in the year 2015. In the case of the electricity, gas and water sector, the total input requirements of the same sector for each \$1 of production are \$ 1.33 in the year 2000 and \$ 1.41 in the year 2015. These are the highest coefficients of the matrix in the two years, which shows that there is a highly productive link within these two sectors but not with the rest of the sectors of the economy, which explains the fact that the indirect effects of the aggregate and speculative oil demand shocks are low in these sectors.

As per the world oil supply, no significant impact on the total economic growth of the country, nor on the production in any sector of the Colombian economy is observed, although most of the sectors show a slight reduction in their productive activity due to increases in world oil production which lower their price. Changes in world oil production inversely affect the behavior of oil prices moderately because variations in a specific moment in the world oil production of some country are usually compensated quickly by variations in the opposite direction in the production of another country. That is, when the world oil supply is contracted due to problems in production or supply by some producer, this production cut is quickly replaced by some other producer. This behavior, which had already been identified by Kilian (2009), implies that the effects on prices due to supply shocks are not strong or sustained, and for this reason changes in world oil production do not significantly affect total or sectoral production in Colombia.

Finally, it is worth mentioning the particular behavior of the mining sector in the face of the three types of shocks. Only the aggregate demand shocks generate a statistically significant impact on the production of the mining sector, and this impact only occurs four years after the shocks occurred. This because higher oil prices encourage

investment in the sector in an oil-exporting economy such as Colombia. However, the results of this investment take a long time before being reflected in oil production. This time, according to the results of this work, is about 4 years or a bit more.

The results obtained in this work are not easily comparable with other studies because, as mentioned in the introduction, there is a little history of studies for emerging oil-exporting economies. Besides, unlike previous empirical literature, this work presents the impacts disaggregated by three types of shocks: supply, aggregate demand and speculative demand for oil.

However, it is worth mentioning that the results obtained in this work differ with those obtained by Francis and Restrepo (2018) for Colombia, who find that industrial production decreases between the second and fifth quarter after receiving the shock, and the Production of the agricultural and mining sectors is not affected. An explanation for the different results found is in the differences in the estimated oil price shocks, given the differences in the methodology for estimating those shocks. In the paper by Francis and Restrepo (2018), an SVAR with information on the Colombian economy is used to estimate oil price shocks, a methodology that is not the most convenient because oil prices are determined based on the interaction of the global supply and demand of oil, which is what is considered in the SVAR estimated in this work as described in Section 2.1.

5. Conclusions

This paper presents an estimate of the effects of shocks in oil prices on total production and by sectors in Colombia. The oil shocks were obtained by estimating a structural VAR model similar to the one proposed by Kilian (2009) with information on world oil production, an index of global economic activity and the real price of oil for a longer period of study. With the estimation of this initial model, it was possible to decompose the changes in oil prices in shocks of aggregate demand, oil supply and specific demand for oil for speculative reasons, finding that the most determining factors in oil prices during the 1994–2017 period were, in order, the specific demand for oil for speculative reasons and global aggregate demand.

Later, to evaluate the effects that these shocks have on the total production in Colombia, another structural VAR model was estimated for the Colombian economy with the total GDP as the only endogenous variable and the oil price shocks obtained in the first model as exogenous variables. It was found that the shocks of specific demand for oil and global aggregate demand are precisely the ones that have the greatest effect on total production in Colombia, as it is found that an increase of 1% in the price of oil which is attributed to growth of global aggregate demand or of specific demand for oil, generates an aggregate increase in aggregate world production of approximately 4.2% and 2%, respectively, after 4 years of the occurrence of shocks. This is expected for an economy in which oil exports become a source of important income for public finances and a driver of national economic activity. On the other hand, oil supply shocks occurred during the study period do not have

important effects on the aggregate of oil production in Colombia, besides not being of magnitude, nor having an important impact on international oil prices.

To determine the sectoral effects of the shocks in oil prices, another VAR model was estimated for each sector in which the production of the sector of interest and the aggregate production of the other sectors of the economy were included as endogenous variables. As a novelty compared to the previous empirical literature, restricted versions of this model are estimated in which the rest of the sectors are “disconnected” assuming that the production variables of these sectors are exogenous. This allows separating the impacts of these shocks between direct and indirect impacts that are transmitted through the other sectors of the economy.

It was found that the shocks of both speculative demand for oil and aggregate demand directly affect mainly the sectors of the manufacturing industry and electricity, gas and water. When the indirect effects transmitted through the other sectors of the economy are also considered, it is found that the construction sector is the most affected positively by the demand shocks. It is followed by the transport, storage and communications sector, and trade, restaurants and hotels sector. The financial intermediation sector receives an important impact in the face of speculative oil demand shocks, but not in the face of aggregate demand shocks. As for the other sectors, the direct and indirect impacts on production in the face of any demand shock are insignificant.

Finally, and consistent with the impacts obtained on the aggregate of production, increases in world oil production that lower oil prices do not generate significant effects on production in any sector of the Colombian economy, although it is worth noting that most of the sectors show a slight reduction in its productive activity and the sectors that receive the greatest direct impacts are again the manufacturing industry and electricity, gas and water.

CRedit authorship contribution statement

Jorge David Quintero Otero: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2020.104691>.

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